2.1.3 Indicators from recurrent processing theory

There are various possible interpretations of RPT that have different implications for AI consciousness. For our purposes, a crucial issue is that the claim that recurrent processing is necessary for consciousness can be interpreted in two different ways. In the brain, it is common for individual neurons to receive inputs that are influenced by their own earlier outputs, as a result of feedback loops from connected regions. However, a form of recurrence can be achieved without this structure: any finite sequence of operations by a network with feedback loops can be mimicked by a suitable feedforward network with enough layers. To achieve this, the feedforward network would have multiple layers with shared weights, so that the same operations would be performed repeatedly—thus mimicking the effect of repeated processing of information by a single set of neurons, which would be produced by a network with feedback loops (Savage 1972, LeCun et al. 2015). In current AI, recurrent neural networks are implemented indistinguishably from deep feedforward networks in which layers share weights, with different groups of input nodes for successive inputs feeding into the network at successive layers.

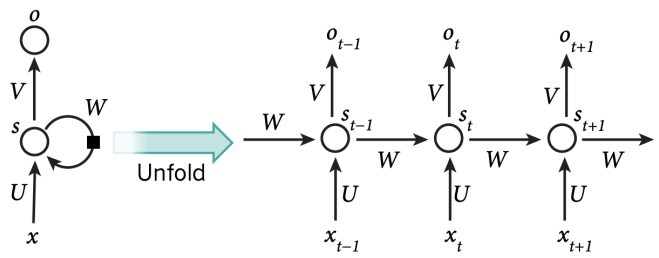

Figure 1: An unfolded recurrent neural network as depicted in LeCun, Bengio, & Hinton (2015). Attribution-Share Alike 4.0 International.

We will say that networks with feedback loops such as those in the brain, which allow individual physically-realised neurons to process information repeatedly, display *implementational recurrence*. However, deep feedforward networks with weight-sharing display only *algorithmic recurrence*—they are algorithmically similar to implementationally recurrent networks but have a different underlying structure. So there are two possible interpretations of RPT available here: it could be interpreted either as claiming that consciousness requires implementational recurrence, or as making only the weaker claim that algorithmic recurrence is required. Doerig et al. (2019) interpret RPT as claiming that implementational recurrence is required for consciousness and criticise it for this claim. However, in personal communication, Lamme has suggested to us that RPT can also be given the weaker algorithmic interpretation.

Implementational and algorithmic recurrence are both possible indicators of consciousness in AI, but we focus on algorithmic recurrence. It is possible to build an artificial system that displays implementational recurrence, but this would involve ensuring that individual neurons were physically realised by specific components in the hardware. This would be a very different approach from standard methods in current AI, in which neural networks are simulated without using specific hardware components to realise each component of the network. An implementational recurrence indicator would therefore be less relevant to our project, so we do not adopt this indicator.

Using algorithmic recurrence, in contrast, is a weak condition that many AI systems already meet. However, it is non-trivial, and we argue below that there are other reasons, besides the

evidence for RPT, to believe that algorithmic recurrence is necessary for consciousness. So we adopt this as our first indicator:

RPT-1: Input modules using algorithmic recurrence

This is an important indicator because systems that lack this feature are significantly worse candidates for consciousness.

RPT also suggests a second indicator, because it may be interpreted as claiming that it is sufficient for consciousness that algorithmic recurrence is used to generate integrated perceptual representations of organised, coherent scenes, with figure-ground segregation and the representation of objects in spatial relations. This second indicator is:

RPT-2: Input modules generating organised, integrated perceptual representations

An important contrast for RPT is between the functions of feature extraction and perceptual organisation. Features in visual scenes can be extracted in unconscious processing in humans, but operations of perceptual organisation such as figure-ground segregation may require conscious vision; this is why RPT-2 stresses organised, integrated perceptual representations.

There are also two further possible interpretations of RPT, which we set aside for different reasons. First, according to the biological interpretation of RPT, recurrent processing in the brain is necessary and sufficient for consciousness because it is associated with certain specific biological phenomena, such as recruiting particular kinds of neurotransmitters and receptors which facilitate synaptic plasticity. This biological interpretation is suggested by some of Lamme's arguments (and was suggested to us by Lamme in personal communication): Lamme (2010) argues that there could be a "fundamental neural difference" between feedforward and recurrent processing in the brain and that we should expect consciousness to be associated with a "basic neural mechanism". We set this interpretation aside because if some particular, biologically-characterised neural mechanism is necessary for consciousness, artificial systems cannot be conscious.

Second, RPT may be understood as a theory only of visual consciousness, which makes no commitments about what is necessary or sufficient for consciousness more generally. On this interpretation, RPT would leave open both: (i) whether non-visual conscious experiences require similar processes to visual ones, and (ii) whether some further background conditions, typically met in humans but not specified by the theory, must be met even for visual consciousness. This interpretation of the theory is reasonable given that the theory has not been extended beyond vision and that it is doubtful whether activity in visual brain areas sustained *in vitro* would be sufficient for consciousness (Block 2005). But on this interpretation, RPT would have very limited implications for AI.

2.2 Global Workspace Theory

2.2.1 Introduction to global workspace theory

The global workspace theory of consciousness (GWT) is founded on the idea that humans and other animals use many specialised systems, often called modules, to perform cognitive tasks

of particular kinds. These specialised systems can perform tasks efficiently, independently and in parallel. However, they are also integrated to form a single system by features of the mind which allow them to share information. This integration makes it possible for modules to operate together in co-ordinated and flexible ways, enhancing the capabilities of the system as a whole. GWT claims that one way in which modules are integrated is by their common access to a "global workspace"—a further "space" in the system where information can be represented. Information represented in the global workspace can influence activity in any of the modules. The workspace has a limited capacity, so an ongoing process of competition and selection is needed to determine what is represented there.

GWT claims that what it is for a state to be conscious is for it to be a representation in the global workspace. Another way to express this claim is that states are conscious when they are "globally broadcast" to many modules, through the workspace. GWT was introduced by Baars (1988) and has been elaborated and defended by Dehaene and colleagues, who have developed a neural version of the theory (Dehaene et al. 1998, 2003, Dehaene & Naccache 2001, Dehaene & Changeux 2011, Mashour et al. 2020). Proponents of GWT argue that the global workspace explains why some privileged subset of perceptual (and other) representations are available at any given time for functions such as reasoning, decision-making and storage in episodic memory. Perceptual representations get stronger due to the strength of the stimulus or are amplified by attention because they are relevant to ongoing tasks; as a result, these representations "win the contest" for entry to the global workspace. This allows them to influence processing in modules other than those that produced them.

The neural version of GWT claims there is a widely distributed network of "workspace neurons", originating in frontoparietal areas, with activity in this network, which is sustained by recurrent processing, constituting conscious representations. When perceptual representations become sufficiently strong, a process called "ignition" takes place in which activity in the workspace neurons comes to code for their content. Ignition is a step-function, so whether a given representation is broadcast, and, therefore, conscious, is not a matter of degree.

GWT is typically presented as a theory of access consciousness—that is, of the phenomenon that some information represented in the brain, but not all, is available for rational decision-making. However, it can also be interpreted as a theory of phenomenal consciousness, motivated by the thought that access consciousness and phenomenal consciousness may coincide, or even be the same property, despite being conceptually distinct (Carruthers 2019). Since our topic is phenomenal consciousness, we interpret the theory in this way. It is notable that although GWT does not explicitly require agency, it can only explain access consciousness if the system is a rational agent since access consciousness is defined as availability for rational control of action (we discuss agency in section 2.4.5)

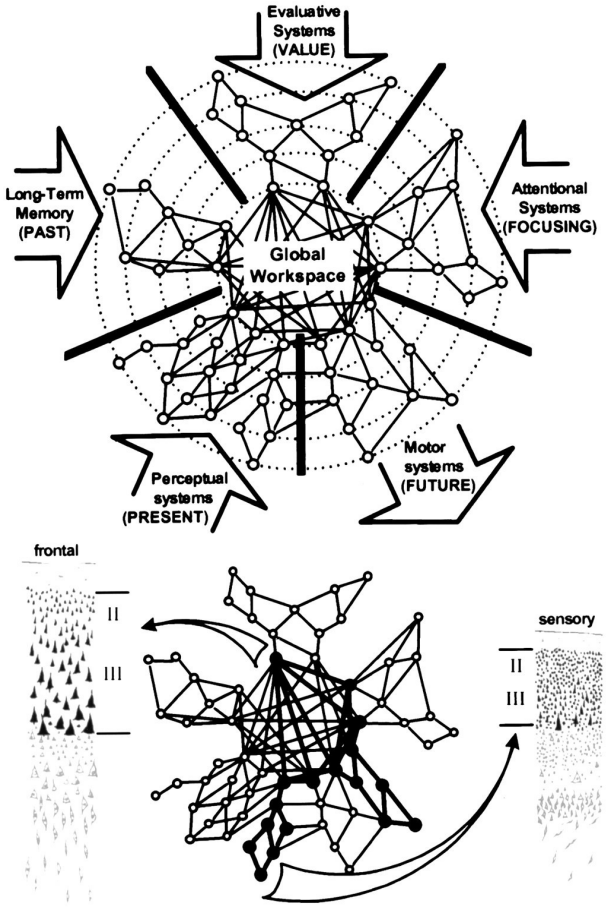

Figure 2: Global Workspace. The figure used by Dehaene et al. (1998) to illustrate the basic idea of a global workspace. Note that broadcast to a wide range of consumer systems such as planning, reasoning and verbal report does not feature in the figure. 1998. National Academy of Sciences. Reprinted with permission.

2.2.2 Evidence for global workspace theory

There is extensive evidence for global workspace theory, drawn from many studies, of which we can mention only a few representative examples (see Dehaene 2014 and Mashour et al. 2020 for reviews). These studies generally employ the method of contrastive analysis, in which brain activity is measured and a comparison is made between conscious and unconscious conditions, with efforts made to control for other differences. Various stimuli and tasks are used to generate the conscious and unconscious conditions, and activity is measured using fMRI, MEG, EEG, or single-cell recordings. According to GWT advocates, these studies show that conscious perception is associated with reverberant activity in widespread networks which include the prefrontal cortex (PFC)—this claim contrasts with the "local" character of RPT discussed above—whereas unconscious states involve more limited activity confined to particular areas. This widespread activity seems to arise late in perceptual processing, around 250-300ms after stimulus onset, supporting the claim that global broadcast requires sustained perceptual representations (Mashour et al. 2020).

Examples of recording studies in monkeys that support a role for PFC in consciousness include experiments by Panagiotaropoulos et al. (2012) and van Vugt et al. (2018). In the former study, researchers were able to decode the presumed content of conscious experience during binocular rivalry from activity in PFC (Panagiotaropoulos et al. 2012). In this study the monkeys viewed stimuli passively—in contrast with many studies supporting GWT—so the results are not confounded by behavioural requirements (this was a no-report paradigm; see sections 1.2.2 and 2.1.2). In the latter, activity was recorded from the visual areas V1 and V4 and dorsolateral PFC, while monkeys performed a task requiring them to respond to weak visual stimuli with eye movements. The monkeys were trained to move their gaze to a default location if they did not see the stimulus and to a different location if they did. Seen stimuli were associated with stronger activity in V1 and V4 and late, substantial activity in PFC. Importantly, while early visual activity registered the objective presence of the stimulus irrespective of the animal's response, PFC activity seemed to encode the conscious percept, as this activity was also present in false alarms—cases in which monkeys acted as though they had seen a stimulus even though no stimulus was present. Activity associated with unseen stimuli tended to be lost in transmission from V1 through V4, to PFC. Studies on humans using different measurement techniques have similarly found that conscious experience is associated with ignition-like activity patterns and decodability from PFC (e.g. Salti et al. 2015).

2.2.3 Indicators from global workspace theory

We want to identify the conditions which must be met for a system to be conscious, according to GWT, because these conditions will be indicators of consciousness in artificial systems. This means that a crucial issue is exactly what it takes for a system to implement a global workspace. Several authors have noted that it is not obvious how similar a system must be to the human mind, in respect of its workspace-like features, to have the kind of global workspace that is sufficient, in context, for consciousness (Bayne 2010, Carruthers 2019, Birch 2022b, Seth & Bayne 2022). There are perhaps four aspects to this problem. First, workspace-like architectures could be used with a variety of different combinations of modules with different capabilities; as Carruthers (2019) points out, humans have a rich and specific set of capabilities that seem to be facilitated by the workspace and may not be shared with other systems. So one question is whether some specific set of modules accessing the workspace is required for workspace activity to be conscious. Second, it's unclear what degree of similarity a process must bear to selection, ignition and broadcasting in the human brain to support consciousness. Third, it is difficult to know what to make of possible systems which use workspace-like mechanisms but in which there are multiple workspaces—perhaps integrating overlapping sets of modules—or in which the workspaces are not global, in the sense that they do not integrate all modules. And fourth, there are arguably two stages involved in global broadcast—selection for representation in the workspace, and uptake by consumer modules—in which case there is a question about which of these makes particular states conscious.

Although these questions are difficult, it is possible that empirical evidence could be brought to bear on them. For example, studies on non-human animals could help to identify a natural kind that includes the human global workspace and facilitates consciousness-linked abilities (Birch 2020). Reflection on AI can also be useful here because we can recognise functional similarities and dis-